NIEM VANNARA

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PROFESSIONAL SUMMARY

Hardworking second-year Data Science and Engineering student with strong foundations in DataScience, Machine Learning, AI, and Python programming. Experienced in data preprocessing, predictive modeling, and real-world dataset analysis. Passionate about applying data-driven solutions to solve practical problems in healthcare, automation, and finance.

EDUCATION

2024 – Present | Bachelor of Data Science and Engineering (Year 2) | Royal University of Phnom Penh

2020-2023 | Finish High School 10 January High School

TECHNICAL PROJECTS

Salary Data Analysis & Exploratory Data Analysis | Python, Pandas, Seaborn, Matplotlib

- Analyzed a real-world salary dataset of **22,770 records** across multiple companies, job roles, and cities in India
- Applied **IQR outlier removal** to normalize salary distribution and improve statistical reliability
- Performed **EDA** to answer 6+ business intelligence questions including top-paying roles, city salary rankings, and employment type impact
- Built a **4-panel analytics dashboard** using Matplotlib subplots visualizing salary distribution, rating correlation, and geographic pay trends
- Computed **Pearson correlation** between company rating and salary, finding a weak positive relationship

AI Face Skin Analysis & Skincare Recommendation | Python, Machine Learning, Computer Vision, Web

- Built an AI-powered web platform that analyzes facial skin conditions via photo or webcam input for the **Huawei ICT Competition Cambodia 2024–2025**
- Developed a computer vision pipeline to detect skin conditions including acne, dryness, oiliness, and sensitivity

- Designed a recommendation engine that maps detected skin conditions to personalized skincare products with match scoring
- Integrated ML model with a responsive web frontend supporting both photo upload and real-time webcam analysis

HR Employee Attrition Analysis & Prediction | Python, Scikit-learn, Pandas, Seaborn, Matplotlib

- Analyzed HR dataset with **35 features** to identify key factors driving employee turnover including income, overtime, age, and job satisfaction
- Preprocessed data using Label Encoding and Z-score StandardScaler with stratified **70/30 train-test split**
- Trained and compared **8 classification models** including Random Forest, SVC, Logistic Regression, KNN, and Decision Tree
- Evaluated models using Accuracy, Precision, Recall, F1 Score, and Confusion Matrix — prioritizing Recall to minimize false negatives
- Applied **GridSearchCV with 5-fold cross-validation** to fine-tune SVC hyperparameters and improve model performance

SKILLS

Data Science & Programming

- Languages: Python (Pandas, NumPy, Scikit-Learn), SQL, C++, HTML/CSS/JavaScript.
- Analysis: Exploratory Data Analysis (EDA), Feature Engineering, Predictive Modeling. Visualization: Matplotlib, Seaborn.

Administrative & Office Management

- Software: MS Office Expert (Word, Excel, PowerPoint).
- Operations: Expert in document management, scheduling, and data confidentiality. Support: Proficient in hardware maintenance and troubleshooting(Printer,Network)

SOFT SKILLS

- Analytical Thinking: Ability to simplify complex data for policy recommendations.
- Organization: Strong time management and accuracy in data entry. Communication: Simplifying technical data for non-technical audiences.

LANGUAGES

- Khmer : Mother Language (Native)
- English : Intermedia

REFERENCES

TOEM THEARA Professor of Royal University of Phnom Penh

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